System-Level Modeling of Multi-Accelerator Architectures via Analytical Cost Model Integration

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ABSTRACT

Modern deep learning workloads are executed on platforms composed of multiple accelerators interconnected through a Network-on-Chip (NoC) and shared memory system. While analytical cost models such as MAESTRO and Timeloop enable efficient performance estimation at individual accelerators, they fail to estimate system-level effects among inter-accelerator communication and shared DRAM access. This project proposes a system-level modeling framework that integrates accelerator-level analytical cost models with explicit modeling of NoC communication and memory behavior. The framework aims to enable performance evaluation and design-space exploration of multi-accelerator systems under realistic workload and mapping scenarios.

1 INTRODUCTION

The rapid growth of deep learning workloads has driven the adoption of specialized hardware accelerators to meet performance and energy-efficiency requirements. Early accelerator designs focused on large monolithic architectures; however, recent trends favor tiled systems composed of multiple accelerator instances connected via a Network-on-Chip (NoC) and shared memory hierarchy. Such architectures improve scalability and flexibility but introduce system-level challenges related to communication, synchronization, and memory contention. This project aims to develop a system-level modeling framework that integrates analytical accelerator cost models with explicit representations of inter-accelerator communication and DRAM access behavior. The resulting framework will enable comparative analysis and optimization of multi-accelerator architectures without resorting to expensive cycle-accurate simulation.

1.1 Motivation

Analytical performance models such as MAESTRO [1] and Timeloop [2] have proven effective for exploring dataflows, memory hierarchies, and mapping strategies for individual accelerators. However, these models primarily focus on single-accelerator execution and do not explicitly account for performance degradation caused by Network-on-Chip (NoC) congestion and shared DRAM contention in multi-accelerator systems.

Prior system-level frameworks, including SCAR [3], MAGMA [4], DREAM [5], and RELMAS [6], emphasize the importance of modeling communication overhead, resource sharing, and scheduling effects across multiple accelerator instances. Nevertheless, these approaches typically rely on their own performance estimation mechanisms rather than leveraging established accelerator-level analytical cost models. This project seeks to unify accelerator-level and system-level modeling to enable more accurate performance estimation and design-space exploration.

2 SPECIFICATION

We consider a tiled multi-accelerator system composed of N accelerator instances connected through a shared Network-on-Chip (NoC) and off-chip DRAM. Each accelerator is modeled using analytical cost models such as Timeloop and MAESTRO [1, 2], which estimate compute latency and local memory behavior for a given DNN layer and mapping.

Building on ideas from MAGMA [4] and SCAR [3], we extend accelerator-level modeling to the system level by incorporating analytical models of:

- NoC bandwidth contention,
- shared DRAM bandwidth allocation,
- concurrent execution of multiple workloads.

The framework estimates end-to-end latency and scalability trends across multiple accelerators. The goal is early-stage design-space exploration rather than cycle-accurate simulation.

2.1 Assessment

The framework will be assessed by comparing the predicted performance trends with existing system-level modeling approaches and the results published in the literature. Evaluation metrics will include execution latency, scalability with respect to the number of accelerator instances, and sensitivity to communication and memory parameters. The goal is not cycle-level accuracy, but performance estimation for early-stage design-space exploration.

2.2 Milestones

(1) Literature review of analytical cost models (MAESTRO, Timeloop) and system-modeling frameworks (MAGMA (up to Section IV) and Section III of SCAR) (by Feb. 18)

(2) Implement output-stationary, weight-stationary, and row-stationary accelerators (ShiDianNao, NVDLA, and Eyeriss) in MAESTRO and Timeloop frameworks and compare their predicted performance. (by Feb. 25)
(3) Study how each accelerator's (ShiDianNao, NVDLA, and Eyeriss) dataflow can be expressed using the framework's semantics (by Feb. 25)
(4) Design the system-level modeling framework (by Mar. 4)
(5) Integration of inter-accelerator communication and DRAM access modeling (by Mar. 10)
(6) Midterm presentation and report (Mar. 11)
(7) Simulation of representative deep learning workloads on multi-accelerator systems (by Apr. 15)
(8) Exploration of system-level optimization strategies inspired by prior work (by May 4)
(9) Final analysis and report preparation (by May 6)
(10) Final presentation and report (May 14)

2.3 Critical Aspects

A key assumption in this project is that accelerator-level analytical cost models, such as MAESTRO and Timeloop, provide sufficiently accurate estimates of compute latency and local memory behavior when applied to individual accelerator instances. Although these models have been validated in prior work, they ignore architectural details and therefore may not capture all hardware effects. At the system level, the modeling of inter-accelerator communication and DRAM access relies on simplified abstractions of NOC traversal and memory contention. These abstractions omit cycle-accurate behavior. As a result, the framework is intended for early-stage design-space exploration rather than precise performance prediction. Ensuring consistency between accelerator-level and system-level assumptions is a critical aspect of the proposed methodology.

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